[인공지능]

Decision Tree

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Tree-based Methods

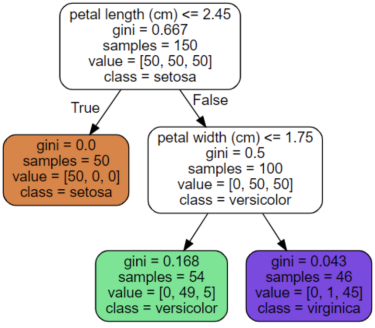
- Tree-based methods involve stratifying or segmenting the predictor space into a number of simple regions.
- Since the set of splitting rules used to segment the predictor space can be summarized in a tree, these types of approaches are known as **decision-tree** methods.
- Decision trees are versatile machine learning algorithms that can perform both classification and regression tasks, and even multi-output tasks.

Pros and Cons of Tree-based Methods

- Tree-based methods are simple and useful for interpretation.
- However, they typically are not competitive with the best supervised learning approaches in terms of prediction accuracy.
- Hence, we also discuss **bagging**, **random forests**, and **boosting**.
  - These methods grow multiple trees which are then combined to yield a single consensus prediction.
- Combining a large number of trees can often result in dramatic improvements in prediction accuracy, at the expense of some loss of interpretation.

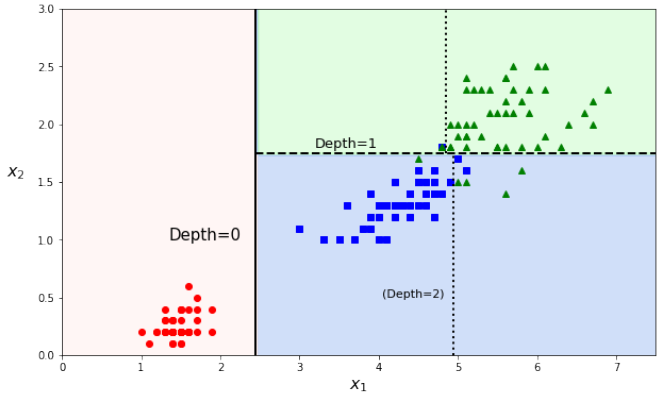
Terminology of Decision Tree

- **Root node**: The topmost node of a tree with depth 0
- **Child node**: A node that descends from another node ($i.e.$, a node connected below a parent)
- **Leaf node**: A node that has no children ($i.e.$, a terminal node)
- Decision trees don't require feature scaling or centering at all.



Iris decision tree

Decision Tree Decision Boundary



Determination of Good Split

- How do we determine a good split?

→ The **Gini's diversity index** is a measure of a node's **impurity**.

→ The **entropy** measure instead by setting the criterion hyperparameter to entropy

→ **Variance (or MSE)** for regression entropy

Gini's Diversity Index

- Gini's diversity index is defined for each tree node i as:

$$G_i = 1 - \sum_{k=1}^{N_{\text{classes}}} \left(\frac{N_{k,i}}{N_i} \right)^2 = 1 - \sum_{k=1}^{N_{\text{classes}}} p_{i,k}^2$$

  - $N_{k,i}$: the number of data samples of class k in node i
  - N_i: the total number of data samples in node i

- The Gini's impurity index ranges from 0 (100% pure node) to $1 - \frac{1}{N_{\text{classes}}}$ (very impure node).

Entropy

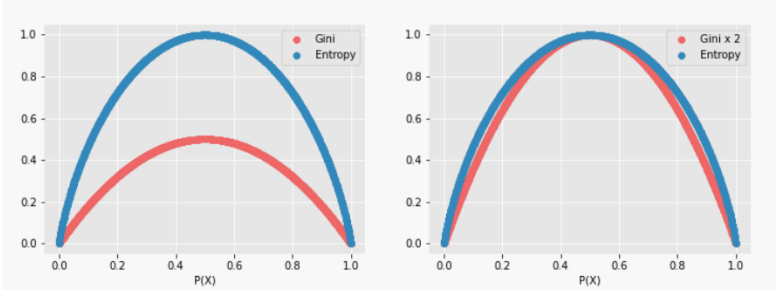
- Entropy measures the uncertainty in a node.
- Entropy is maximized when all classes are equally likely and equals zero when the node contains only one class.

$$H_i = - \sum_{\substack{k=1 \\ p_{i,k} \neq 0}}^{N_{\text{classes}}} p_{i,k} \log(p_{i,k})$$

  - $p_{i,k}$: proportion of class k in node i
  - N_{classes}: number of classes
  - If the node contains only one class, certainty $\uparrow \Rightarrow$ entropy value $= 0$.
  - If the classes are evenly mixed, uncertainty $\uparrow \Rightarrow$ entropy value $\uparrow$.

Gini's Index vs. Entropy

- Gini's diversity index slightly faster, strong default choice.
- Entropy can react more to minority class.



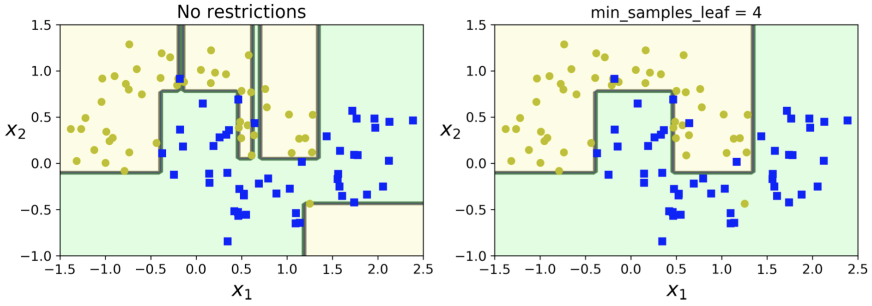
Source from Quantdare

[Pre-pruning] Regularization Hyperparameters

- Decision trees make very few assumptions about the training data (as opposed to linear models, which obviously assume that the data is linear, for example).
- If left unconstrained, the tree structure will adapt itself to the training data, fitting it very closely, and most likely overfitting it.
- To avoid overfitting the training data, you need to restrict the decision tree's freedom during training.

→ Reducing max depth will regularize the model and thus reduce the risk of overfitting.

  - *e.g.*, the minimum number of samples a node must have before node can be split, the minimum number of samples a leaf node must have, etc.



Regularization using min samples leaf

Example: Gini Index

- CART (Classification and Regression Trees) uses Gini impurity for classification and selects the binary split with the largest impurity reduction.
- 7 customers: four input features and a binary purchase label.

ID	Age (x_{i1})	Income (x_{i2})	Student Status (x_{i3})	Credit Rating (x_{i4})	Purchase (y_i)
1	Adolescent	High	No	Good	No Purchase
2	Adolescent	High	No	Excellent	No Purchase
3	Young Adult	High	No	Good	Purchase
4	Middle-aged	Medium	No	Good	Purchase
5	Middle-aged	Low	Yes	Good	Purchase
6	Middle-aged	Low	Yes	Excellent	No Purchase
7	Young Adult	Low	Yes	Excellent	Purchase

- Parent node D: 4 buyers and 3 non-buyers.
  - Class proportions: $p_{\text{Yes}} = 4/7$ and $p_{\text{No}} = 3/7$.

$$G(D) = 1 - \left(\frac{4}{7}\right)^2 - \left(\frac{3}{7}\right)^2 = \frac{24}{49} \approx \boxed{0.490}.$$

- Impurity reduction measures the decrease in Gini impurity:

$$\Delta G(D, s) = G(D) - \Bigg(\underbrace{\frac{|D_L|}{|D|} G(D_L) + \frac{|D_R|}{|D|} G(D_R)}_{\text{Weighted Gini impurity after splitting}} \Bigg).$$

Here, s is a candidate binary split, and D_L and D_R are the resulting child datasets.

- The split with the largest impurity reduction is selected.

Age Split: Adolescent vs. Others

- Left child: Adolescent
  - 0 buyers and 2 non-buyers.
- Right child: Young Adult and Middle-aged
  - 4 buyers and 1 non-buyer.

$$G(D_L) = 1 - \left(\frac{0}{2}\right)^2 - \left(\frac{2}{2}\right)^2 = 0, \qquad G(D_R) = 1 - \left(\frac{4}{5}\right)^2 - \left(\frac{1}{5}\right)^2 = 0.320.$$

$$\begin{aligned} G_{\text{split}}(D, s_1) &= \frac{2}{7} G(D_L) + \frac{5}{7} G(D_R) \\ &= \frac{2}{7}(0) + \frac{5}{7}(0.320) \\ &\approx 0.229. \end{aligned}$$

$$\Delta G(D, s_1) = 0.49 - 0.23 \approx \boxed{0.26}.$$

Age Split: Adolescent and Young Adult vs. Middle-aged

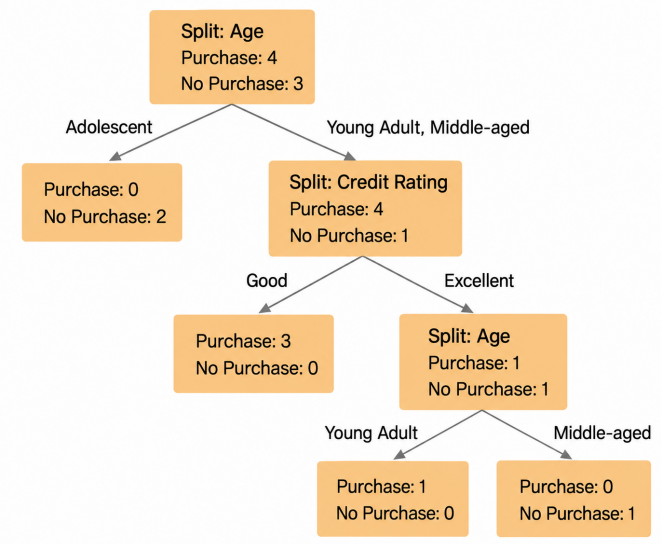
- Left child: Adolescent and Young Adult
  - 2 buyers and 2 non-buyers.
- Right child: Middle-aged
  - 2 buyers and 1 non-buyer.

$$G(D_L) = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 = \frac{1}{2}, \qquad G(D_R) = 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = \frac{4}{9}.$$

$$\begin{aligned} G_{\text{split}}(D, s_2) &= \frac{4}{7} G(D_L) + \frac{3}{7} G(D_R) \\ &= \frac{4}{7}\left(\frac{1}{2}\right) + \frac{3}{7}\left(\frac{4}{9}\right) \\ &= \frac{10}{21} \approx 0.476. \end{aligned}$$

$$\Delta G(D, s_2) = 0.49 - 0.476 \approx \boxed{0.014}.$$

- Repeat the same process for Income, Student Status, and Credit Rating.



Example: Entropy

- ID3 (Iterative Dichotomiser 3) uses entropy to calculate information gain and selects the attribute with the highest information gain.
- 14 customers: four input features and a binary purchase label.

ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Young adult	High	No	Fair	Yes
4	Middle-aged	Medium	No	Fair	Yes
5	Middle-aged	Low	Yes	Fair	Yes
6	Middle-aged	Low	Yes	Excellent	No
7	Young adult	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Middle-aged	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Young adult	Medium	No	Excellent	Yes
13	Young adult	High	Yes	Fair	Yes
14	Middle-aged	Medium	No	Excellent	No

- Parent node D: 9 buyers and 5 non-buyers.
  - Class proportions: $p_{\text{Yes}} = 9/14$ and $p_{\text{No}} = 5/14$.

$$H(D) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} \approx \boxed{0.940}.$$

- Information gain measures the reduction in entropy:

$$\text{Gain}(D, A) = H(D) - \sum_{v \in \mathcal{V}_A} \frac{|D_v|}{|D|} H(D_v),$$

where A is a candidate feature, $\mathcal{V}_A$ is its set of categories, and D_v contains the samples with $A = v$.

- Information gain for age
  - Youth, young adult, middle-aged

$$\text{Gain}(D, age) = H(D) - H(D_{age})$$

$$\begin{aligned} H(D_{age}) = & \frac{5}{14}\left(-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}\right) \\ & + \frac{4}{14}\left(-\frac{4}{4}\log_2\frac{4}{4}\right) \\ & + \frac{5}{14}\left(-\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}\right) \\ = & 0.694 \end{aligned}$$

$$\text{Gain}(D, age) = 0.94 - 0.694 = 0.246$$

- Information gain for income
  - High, medium, low

$$\text{Gain}(D, income) = H(D) - H(D_{income})$$

$$\begin{aligned} H(D_{income}) = & \frac{4}{14}\left(-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right) \\ & + \frac{6}{14}\left(-\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6}\right) \\ & + \frac{4}{14}\left(-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}\right) \\ = & \ 0.911 \end{aligned}$$

$$\text{Gain}(D, income) = 0.94 - 0.911 = 0.029$$

- Information gain for student status
  - Yes, no

$$\text{Gain}(D, student) = H(D) - H(D_{student})$$

$$\begin{aligned} H(D_{student}) &= \frac{7}{14}\left(-\frac{6}{7}\log_2\frac{6}{7} - \frac{1}{7}\log_2\frac{1}{7}\right) \\ &\quad + \frac{7}{14}\left(-\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7}\right) \\ &= 0.789 \end{aligned}$$

$$\text{Gain}(D, student) = 0.94 - 0.789 = 0.151$$

- Information gain for credit rating
  - Excellent, fair

$$\text{Gain}(D, credit) = H(D) - H(D_{credit})$$

$$\begin{aligned} H(D_{credit}) &= \frac{8}{14}\left(-\frac{6}{8}\log_2\frac{6}{8} - \frac{2}{8}\log_2\frac{2}{8}\right) \\ &\quad + \frac{6}{14}\left(-\frac{3}{6}\log_2\frac{3}{6} - \frac{3}{6}\log_2\frac{3}{6}\right) \\ &= 0.892 \end{aligned}$$

$$\text{Gain}(D, credit) = 0.94 - 0.892 = 0.048$$

- Select the attribute with the highest information gain as the splitting attribute.

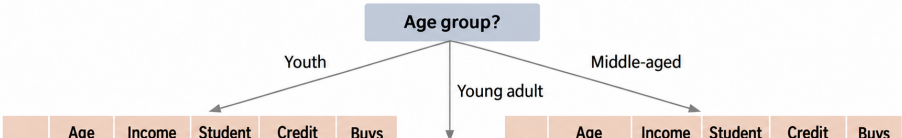
$$\text{Gain}(D, age) = H(D) - H(D_{age}) = 0.247$$

$$\text{Gain}(D, income) = H(D) - H(D_{income}) = 0.029$$

$$\text{Gain}(D, student) = H(D) - H(D_{student}) = 0.151$$

$$\text{Gain}(D, credit) = H(D) - H(D_{credit}) = 0.048$$

  - Age has the highest information gain of 0.247.
  - The ID3 algorithm selects Age as the first splitting attribute.
  - All samples in the young adult group are classified as Purchase, so no further splitting is needed.
  - The adolescent and middle-aged groups contain both Purchase and No Purchase samples, so further splitting is required.



ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes

ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
4	Middle-aged	Medium	No	Fair	Yes
5	Middle-aged	Low	Yes	Fair	Yes
6	Middle-aged	Low	Yes	Excellent	No
10	Middle-aged	Medium	Yes	Fair	Yes
14	Middle-aged	Medium	No	Excellent	No

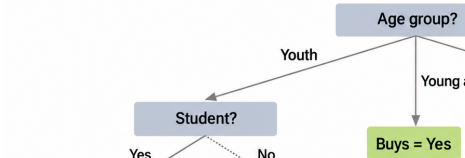
- Calculate the information gain for Income, Student Status, and Credit Rating within the youth group.

$$H(D_{youth-income}) = -\left\{ \frac{1}{5}\left(\frac{1}{1}\log_2\frac{1}{1}\right) + \frac{2}{5}\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) + \frac{2}{5}\left(\frac{2}{2}\log_2\frac{2}{2}\right)\right\} = 0.4$$

$$H(D_{youth-student}) = -\left\{ \frac{2}{5}\left(\frac{2}{2}\log_2\frac{2}{2}\right) + \frac{3}{5}\left(\frac{3}{3}\log_2\frac{3}{3}\right)\right\} = 0$$

$$H(D_{youth-credit}) = -\left\{ \frac{2}{5}\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) + \frac{3}{5}\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right)\right\} = 0.951$$

- Within the adolescent group, Student Status has the highest information gain and is selected as the second splitting attribute.



ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes

ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
4	Middle-aged	Medium	No	Fair	Yes
5	Middle-aged	Low	Yes	Fair	Yes
6	Middle-aged	Low	Yes	Excellent	No
10	Middle-aged	Medium	Yes	Fair	Yes
14	Middle-aged	Medium	No	Excellent	No

- Calculate the information gain for Income, Student Status, and Credit Rating within the middle-aged group as well.

$$Gain(D_{middel-aged}, income) = H(D_{middel-aged}) - H(D_{middel-aged-income}) = 0.97 - 0.95 = 0.02$$

$$Gain(D_{middel-aged}, student) = H(D_{middel-aged}) - H(D_{middel-aged-student}) = 0.97 - 0.95 = 0.02$$

$$Gain(D_{middel-aged}, credit) = H(D_{middel-aged}) - H(D_{middel-aged-credit}) = 0.971 - 0 = 0.971$$

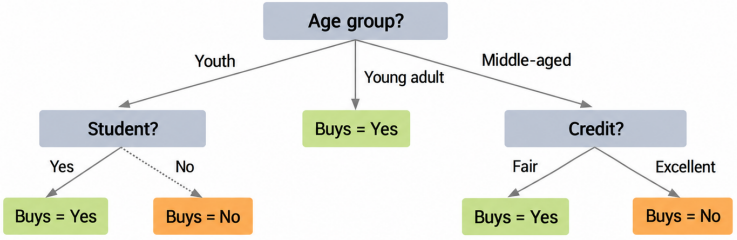
$$H(D_{middel-aged}) = -\left(\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right) = 0.971$$

$$H(D_{middel-aged-income}) = -\left\{ \frac{2}{5}\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) + \frac{3}{5}\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) \right\} = 0.951$$

$$H(D_{middel-aged-student}) = -\left\{ \frac{2}{5}\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) + \frac{3}{5}\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) \right\} = 0.951$$

$$H(D_{middel-aged-credit}) = -\left\{ \frac{2}{5}\left(\frac{2}{2}\log_2\frac{2}{2}\right) + \frac{3}{5}\left(\frac{3}{3}\log_2\frac{3}{3}\right) \right\} = 0$$

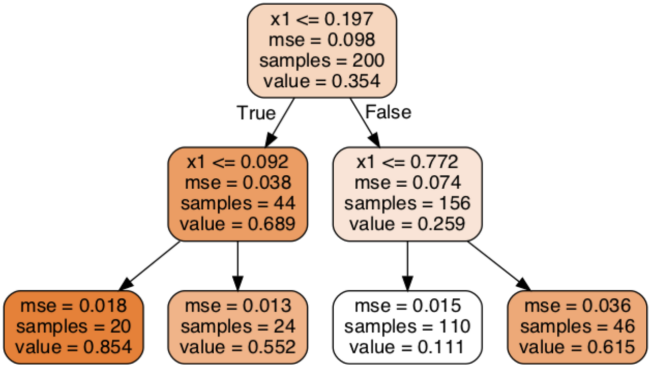
- Within the middle-aged group, Credit Rating has the highest information gain and is selected as the second splitting attribute.



ID	Age (x_{i1})	Income (x_{i2})	Student (x_{i3})	Credit (x_{i4})	Buys (y_i)
4	Middle-aged	Medium	No	Fair	Yes
5	Middle-aged	Low	Yes	Fair	Yes
6	Middle-aged	Low	Yes	Excellent	No
10	Middle-aged	Medium	Yes	Fair	Yes
14	Middle-aged	Medium	No	Excellent	No

Decision Tree Regression

- Decision trees are also capable of performing regression tasks.



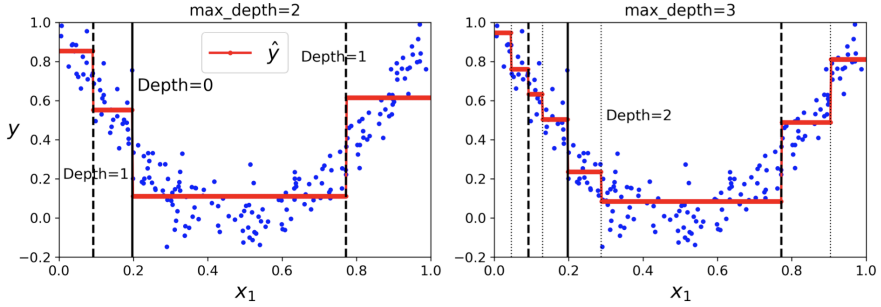
A decision tree for regression

The CART Training Algorithm for Regression

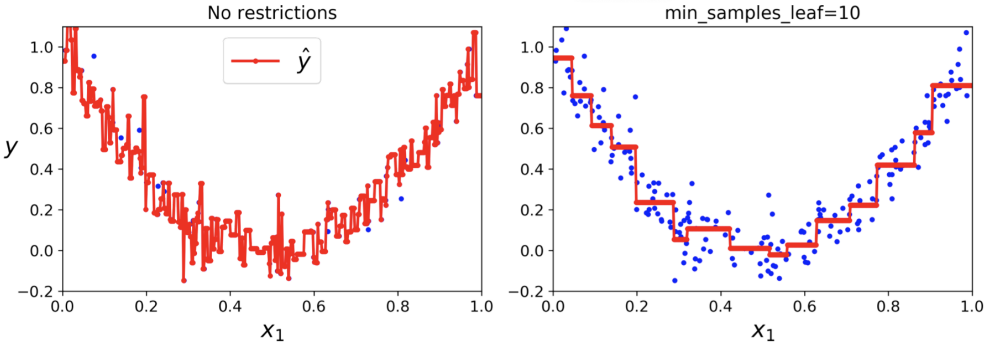
- CART cost function for regression
  - Instead of trying to split the training set in a way that minimizes impurity, it now tries to split the training set in a way that minimizes the MSE (Mean Squared Error).

$$J(k, t_k) = \frac{m_{\text{left}}}{m}\text{MSE}_{\text{left}} + \frac{m_{\text{right}}}{m}\text{MSE}_{\text{right}} \quad \text{where} \quad \begin{cases} \text{MSE}_{\text{node}} = \sum\limits_{i \in \text{node}} \left(\hat{y}_{\text{node}} - y^{(i)}\right)^2 \\ \hat{y}_{\text{node}} = \dfrac{1}{m_{\text{node}}} \sum\limits_{i \in \text{node}} y^{(i)} \end{cases}$$

Predictions of Decision Tree for Regression



Prediction of two decision tree regression models

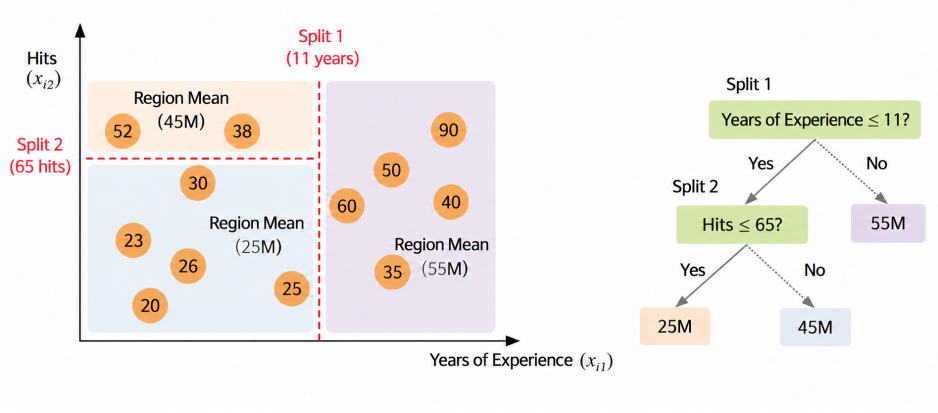


Regularizing decision tree regression

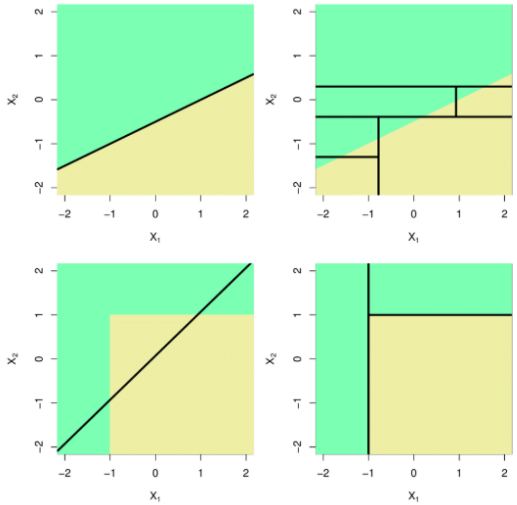
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Example

- Using the CART Regression Algorithm, we can predict a baseball player's salary based on various features.



Linear vs Tree Model



Limitations of Decision Tree

- Decision trees are simple to understand and interpret, easy to use, versatile, and powerful.
- However, they do have a few limitations.
  - Decision trees love orthogonal decision boundaries (all splits are perpendicular to an axis), which makes them sensitive to training set rotation.
  - Decision trees are very sensitive to small variations in the training data.

Thank You!

(Q & A)

References

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- 권현 외. (2024) 데이터, 모델, 학습이론으로 배우는 머신러닝 이론과 실습. 한빛아카데미.